Mazen Ashraf Refaat

Game Developer | XR & Simulation | Unity | Real-Time 3D

Nasr City | mazenashraframzy@gmail.com | 01159034434 | linkedin.com/in/mazen-ashraf-73b601277
github.com/MazenAshrafRamzi | **Portfolio (playable demos):** mazenashraframzi.github.io

Summary

Game and XR Developer with a Computer Science degree, specializing in real-time 3D applications including VR/AR simulations, interactive training systems, educational game suites and interactive brand pitches. Primary engine expertise in Unity (C#, URP, OpenXR, AR Foundation, XR Interaction Toolkit) with shipped WebGL, Android and Windows builds. Published researcher (Springer CCIS, PAAMS 2025) with hands-on experience in photogrammetry, AI-assisted 3D asset pipelines, procedural animation and event-driven gameplay architecture. Seeking Game Developer, XR Developer, or Simulation Developer roles across real-time 3D platforms.

Education

German University in Cairo, Bachelor of Science in Computer Science and Engineering — Sept 2020 – July 2025

- GPA: Very Good
- **Relevant Coursework:** Computer Graphics, VR and AR Development, Advanced Unity Lab, Computer Vision, Artificial Intelligence, Machine Learning, Software Engineering, Data Structures and Algorithms, Cloud Computing, Network Security, Database Management Systems

Sahara International School

- GPA: 97% (IGCSE System)

Core Skills

Unity, C#, URP, OpenXR, AR Foundation, Unity XR Interaction Toolkit, VR Development, AR Development, WebGL Builds, 3D Simulation, Game Development, Procedural Animation & IK Rigs, DOTween, Blender, Photogrammetry, ShaderGraph, uGUI / UI Toolkit, Arabic / RTL UI, Performance Optimization, Mobile Optimization, Event-Driven Architecture, Git, Python, Computer Vision, Agile / Scrum, Plastic SCM, AI 3D Generation (Tripo3D.ai, Meshy.ai), Mixamo, ElevenLabs

Work Experience

Al Motahida Education, Unity Developer — May 2026 – Present

- *Interactive Mini-Games & Educational Gameplay Systems*: contributing to the development and expansion of a modular suite of 2D educational mini-games and interactive exercises integrated into digital eBook curricula for the **Ministry of Education, Sultanate of Oman**, comprising **15+ exercises** across various subjects.
- Extended and maintained **diverse gameplay interaction mechanics** (drag-and-drop, item sorting, multi-selection, touch gestures and canvas drawing), adding DOTween animations so exercises feel responsive and polished.
- Implemented **sorting mechanics** by extending the existing drag-and-drop system, adapting the interaction model to new exercise requirements within the established architecture.
- Worked within an **event-driven sequence control architecture** to integrate new educational exercises, managing game flows, state transitions and step navigation across multiple modules.
- Integrated **scene state persistence** and runtime snapshot systems into new exercises, so player progress and scoring data survive scene reloads.
- Contributed to **real-time feedback pipelines** by hooking in progress bars, tween animations and synchronized SFX queues across newly added educational content.
- Applied **WebGL-specific optimizations** and texture compression to cut load times and runtime memory, meeting the performance limits of exercises embedded in digital eBook curricula.
- Integrated the team's **RTL text handling system** across Arabic-language exercises so text renders and lays out in the correct direction throughout.
- Used **custom Unity Editor tools** and Odin Inspector extensions to set up assets and integrate level content efficiently within the team's Git-based workflow.
- **Tools:** C#, Unity, DOTween, Odin Inspector, JSON, Git

7P Software and Marketing, Unity Developer January 2026 – May 2026

- *Automotive Testing Training Simulation*: a 3D interactive training tool for vehicle diagnostics.
- Built the full interactive simulation scene: live **diagnostic screen panels** driven by real-time scene state, mirroring the physical device's behavior.
- Engineered a **component simulation system** for resistors, capacitors, fuses and MOSFETs, each with its own values, types and orientations, routing the correct readings and feedback states for each component's configuration.
- Built a **Multimeter simulation** whose display changes with the selected measurement mode (voltage, capacitance, etc.). It reads the connected component's value and checks probe orientation, forming the core assessment mechanic of Level 1.
- Simulated a **high-complexity automotive tester device** with ~30 connection points, mirroring how a real tester responds to different components, values and wiring configurations across diagnostic scenarios.
- Designed a **tutorial level** that guides users through electrical circuit exercises before they move on to automotive-specific diagnostics.
- Implemented a **user authentication system** (Sign Up, Login, Password Change) with local data persistence and clean backend integration points.
- Built a **level selection system** with per-level and global progress bars; sequential unlock logic enforces a structured training path.
- Sourced and processed **10+ 3D assets** through AI generation (Tripo3D.ai) and photogrammetry (RealityScan), optimizing each in Blender for real-time performance.
- **Tools:** C#, Unity, Blender, Tripo3D.ai, RealityScan

Recent Projects (2026) NEW

Voices of Stone: Interactive Museum of Ancient Egypt Play in browser

- Built a Unity 6 (URP) museum demo for **Windows, Android and WebGL** in which three statues (Ra-Horakhty, Bastet, Anubis) narrate their lore in the first person, with cinematic camera cuts to each detail described, tappable hotspots, fact cards and a timeline of Egyptian periods.
- Prepared client **photogrammetry scans** in Blender: decimated a 2.9M-triangle scan to 459k with UDIM textures preserved, and derived normal and metal maps for gilded surfaces.
- Delivered a full **English / Arabic (RTL)** mode with Egyptian neural voices and live language switching, three lighting moods (including a moving sun and flickering torchlight), and procedural ambient music.
- **Tools:** Unity, URP, C#, Blender, Photogrammetry, Neural TTS, WebGL

Ziggy Burger Lab & Menu Lab (Tajriba's Menu): Interactive 3D Ordering Experiences Play in browser

- Built a **WebGL burger builder** for a restaurant brand: a tablet-style ordering UI that flows into a 3D build view where users stack, inspect, explode and order their burger. It uses procedural food meshes, a custom camera rig, in-house tween and spring systems (no third-party packages) and synthesized, baked audio.
- Generalized it into **Menu Lab**, a white-label 3D restaurant menu driven by JSON presets (branding, fonts, prices) and menu data, so a new client needs no rebuild. It includes day/night themes, dish customization, cart and receipt flow, and English / Arabic UI.
- **Tools:** Unity, URP, C#, Input System, TextMeshPro, WebGL

Bad Ziggy: Interactive 9:16 Brand Pitch

- Developed a ~90 s playable story across 11 beats (skateboarding, shop visit, nostalgia world, finale), playable by touch or as a self-running demo, with a full English / Arabic toggle.
- Built a cut-out character with an **IK rig and a procedural animation layer** (breathing, sway, follow-through), a gesture and cue-feedback system, and a texel-space grain shader for shimmer-free scrolling.
- **Tools:** Unity 2D, C#, Custom Shaders, Procedural Animation

Perfect Human: Cartoon Muhammad Ali Interactive Pitch Watch

- Created a hands-on **character creator** (slider, drag, swipe and tap interactions synced to client voice-over) that transforms a mascot into a cartoon Muhammad Ali, with an inspect mode for orbiting, posing, expressions and boxing moves. Shipped for Windows and Android.
- Built all audio procedurally with a NaN-safe master limiter, and verified the full interactive run with automated Input System simulation.
- **Tools:** Unity, URP, C#, Toon Shading, Input System, Procedural Audio

Limescents: "Do You Smell That?" Motion-Graphics Promo Watch

- Directed and produced a 35 s, 1080×1920 agency launch film rendered entirely from code, cut to a 96 bpm track, built around a single animated brand element that drives every transition. Wrote the treatment and structure comparison.
- **Tools:** JavaScript, Code-Rendered Animation, ffmpeg, Sound Design

Projects

Virtual Reality Time Travel Museum GitHub

- Built a **collaborative VR museum** in Unity (OpenXR) with interactive environments, spatial audio and narrative-driven exploration of Roman history.
- Reconstructed **Roman statues via photogrammetry** and integrated the optimized models as interactive centrepiece exhibits.
- Developed an **AI-driven NPC** using Meshy.ai 3D generation, Mixamo animations and ElevenLabs synthetic voice lines.
- **Tools:** Unity, OpenXR, Photogrammetry, Meshy.ai, Mixamo, ElevenLabs

Control-Inspired Third-Person Shooter

- Designed **4 distinct enemy types** (armored, levitating, invisible, explosive), each with its own AI behaviors and Mixamo animations.
- Built a **boss fight** with an arena layout, projectile attack patterns, falling tile hazards and headshot-only defeat mechanics.
- **Tools:** Unity, Unity Version Control (Plastic SCM)

Bachelor Project: 3D Model Generation for XR Training

- Converted **orthographic engineering drawings** into XR-ready 3D models (OBJ) through automated Blender extrusion workflows.
- Applied **computer vision techniques** (edge detection, contour analysis) to extract features from engineering input images.
- **Published** as a conference paper: *PAAMS 2025, Springer CCIS*, a peer-reviewed international conference.
- **Tools:** Blender, Python, Computer Vision

Augmented Reality Pirate Treasure Hunt (Mobile) GitHub

- Designed an **AR mobile game** (AR Foundation) with real-world plane detection, NPC interaction and touch-based puzzle mechanics.
- Guided players with **AR particle effects** on detected planes, and built animated victory sequences with character interaction.
- **Tools:** Unity, AR Foundation, Mixamo, ElevenLabs, AI Voice Tools

Extra-Curricular Activities

Model of European Union – Ain Shams University 2 years

- Served as junior delegate, senior delegate and teaching secretariat (2020); presented on the history and EU-funded applications of Virtual Reality technology.

Visual Game Studio Club – German University in Cairo 2022

- Studied game design fundamentals and developed projects using Unity and C# as a junior member.

Technical Skills

Languages: C#, Python, C++, C, Java, JavaScript, SQL, HTML, CSS
XR & Game Dev: Unity, URP, OpenXR, AR Foundation, Unity XR Interaction Toolkit, Input System, DOTween, ShaderGraph, uGUI / UI Toolkit, TextMeshPro, Mixamo, Plastic SCM, WebGL / Android / Windows builds
3D & Asset Pipeline: Blender, Photogrammetry (RealityScan), AI 3D Generation (Tripo3D.ai, Meshy.ai), ElevenLabs & Neural TTS (AI voice), OpenGL
Engineering Tools: Git, GitHub, Visual Studio, VS Code, Docker, Kubernetes, npm, React, Node.js, ffmpeg, Render
Databases: MongoDB, MySQL, PostgreSQL (Neon)
Practices: Agile / Scrum, Event-Driven Architecture, Performance Optimization, Mobile Optimization, Computer Vision